FM Bandwidth Calculations

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Frequency Spectrum of FM Wave

$$v_{fm}(t) = V_c \sin[\omega_c t + m_f \sin\omega_m t]$$

Apply Bessel's function $J_0, J_1, J_2 \cdots J_3$

A_m

$\downarrow$

$\frac{m V_c}{2}$

Bessel function constant

$$v_{fm}(t) = \{ \underbrace{V_c J_0(m_f) \sin\omega_c t}_{\text{carrier}}$$

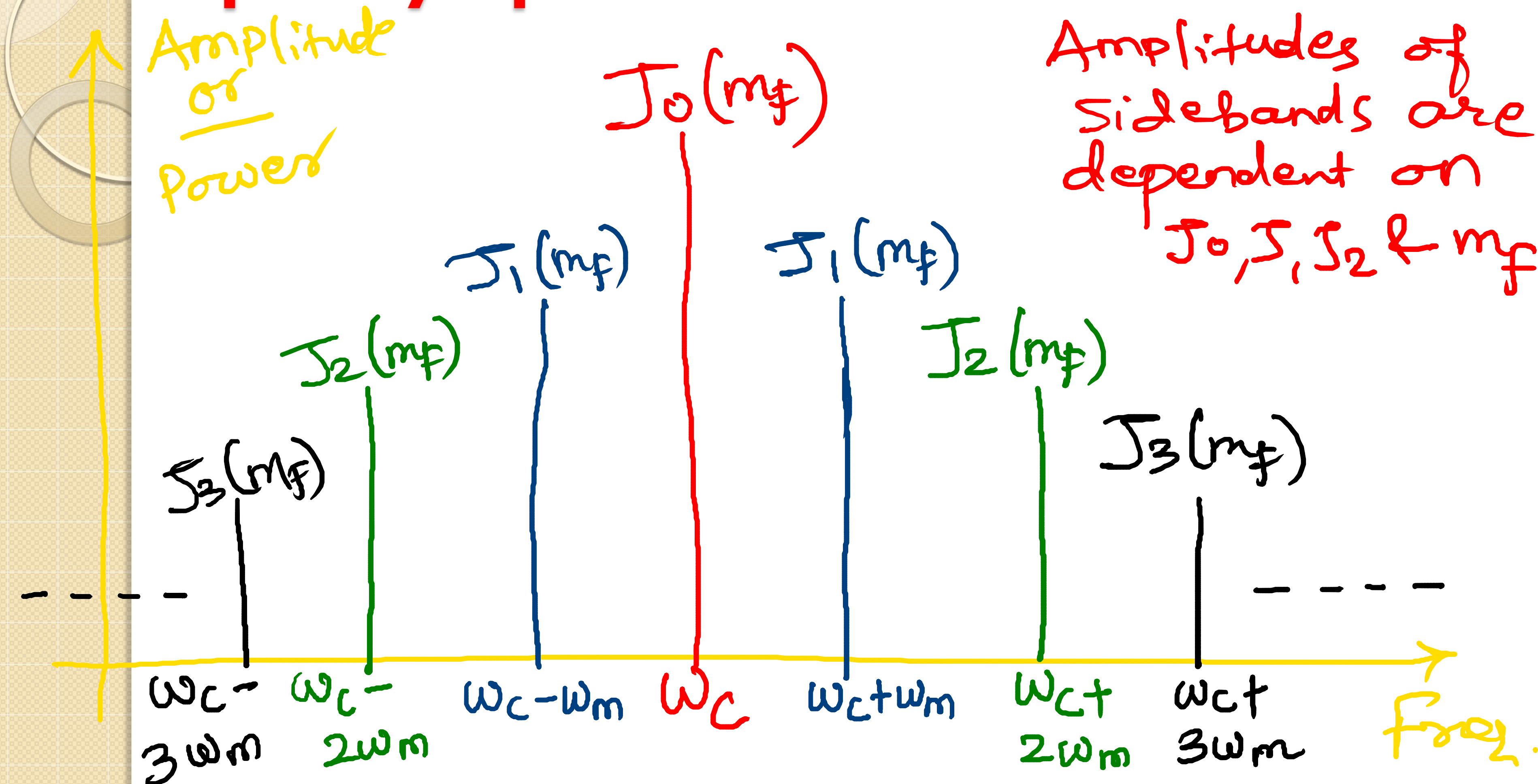
$$+ J_1(m_f)[\underbrace{\sin(\omega_c + \omega_m)t}_{} - \underbrace{\sin(\omega_c - \omega_m)t}_{\text{1st side band}}$$

$$+ J_2(m_f)[\sin(\omega_c + 2\omega_m)t + \sin(\omega_c - 2\omega_m)t \quad \text{2nd side band}$$

$$+ J_3(m_f)[\sin(\omega_c + 3\omega_m)t - \sin(\omega_c - 3\omega_m)t \quad \text{3rd sideband}$$

$$+ \cdots\cdots J_n(m_f) \cdots \}$$

Frequency Spectrum of FM Wave



Amplitudes of sidebands are dependent on J_0, J_1, J_2 & m_f

Frequency Spectrum of FM Wave

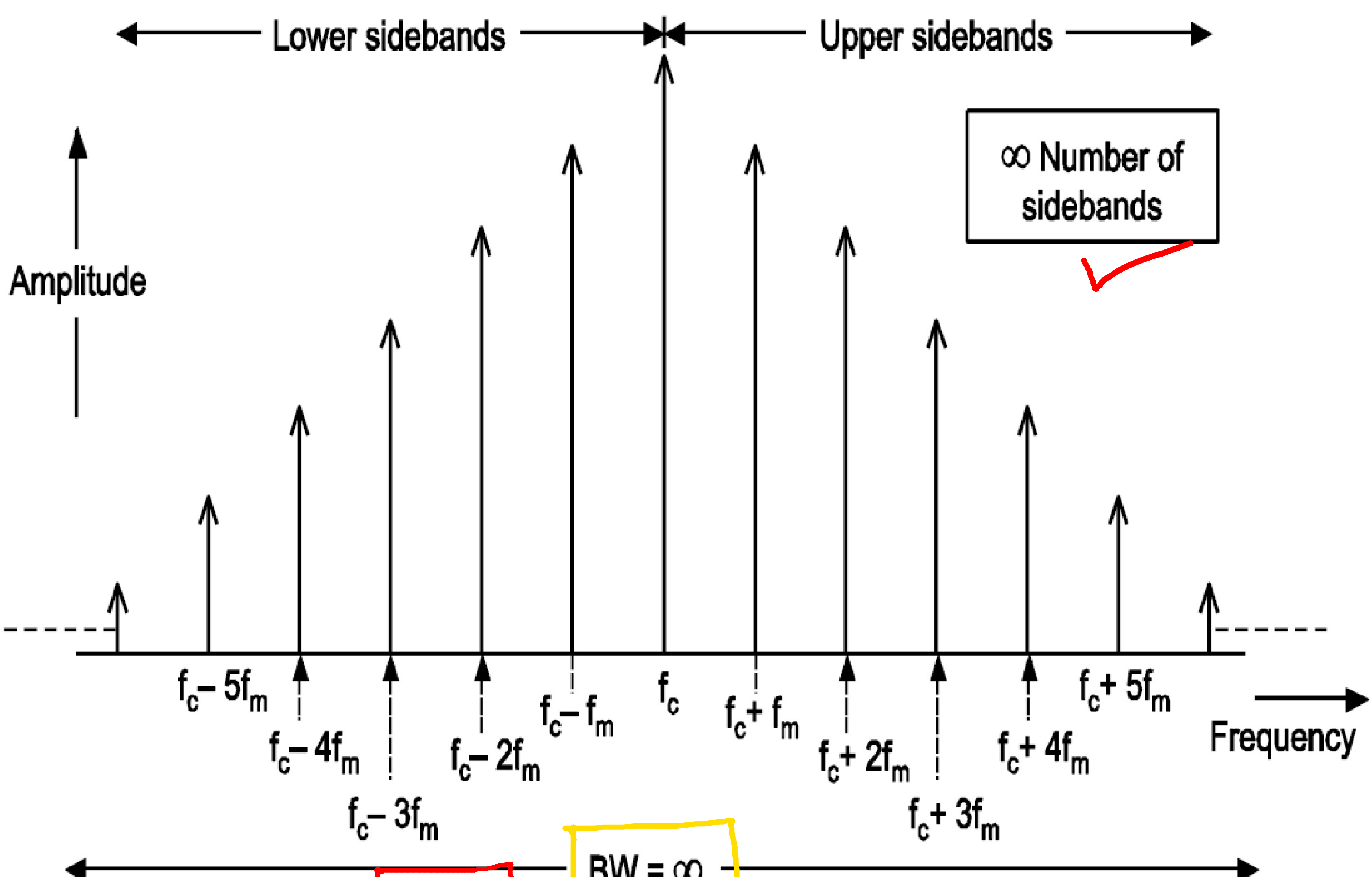


Fig. : Ideal Frequency Spectrum of FM

Bessel Function Table

J_0 J_1 J_2 J_3 J_4 J_{16}

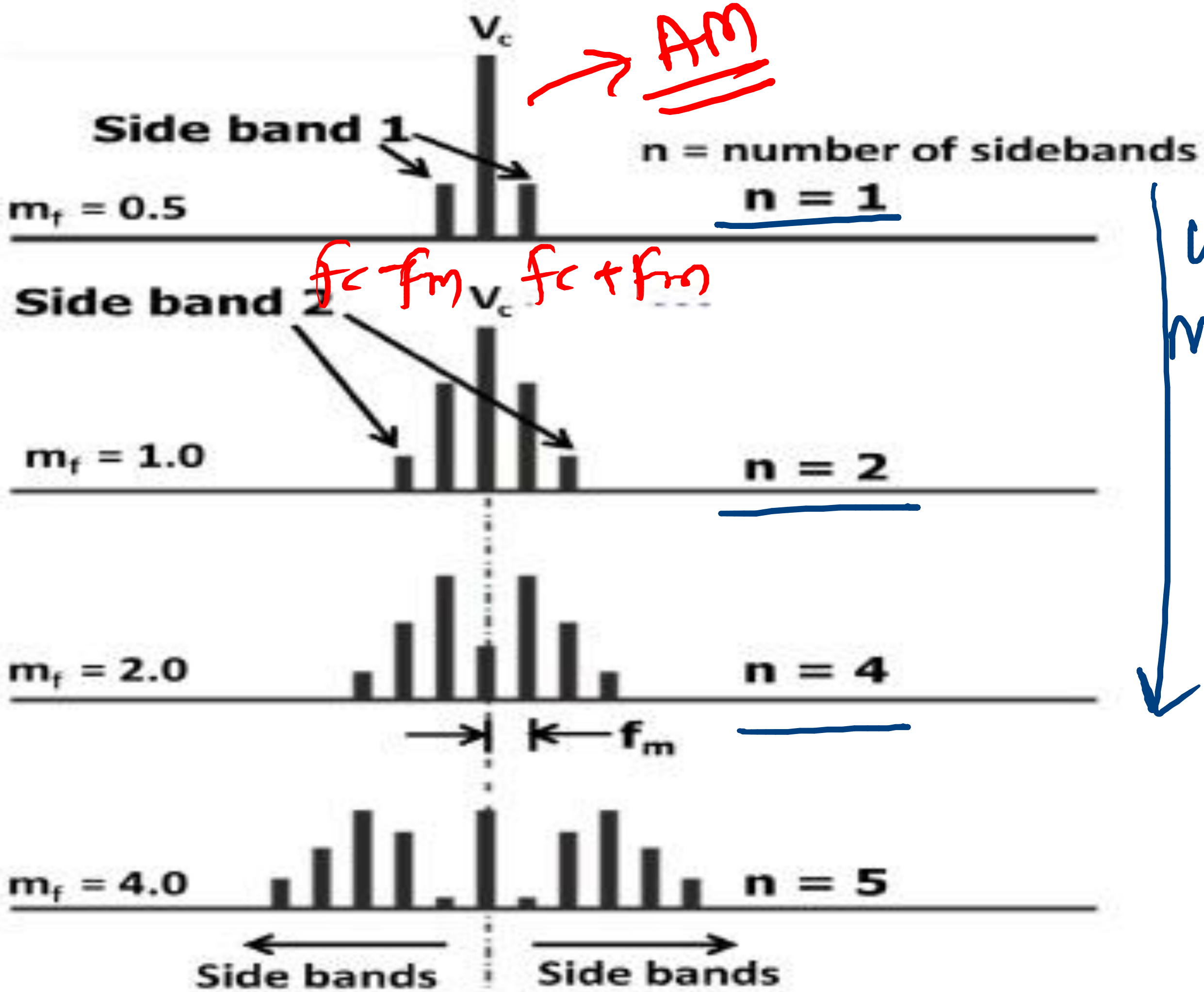
Modulation index	Carrier	Sideband															
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0.00	1.00																
0.25	0.98	0.12															
0.5	0.94	0.24	0.03														
1.0	0.77	0.44	0.11	0.02													
1.5	0.51	0.56	0.23	0.06	0.01												
2.0	0.22	0.58	0.35	0.13	0.03												
2.41	0	0.52	0.43	0.20	0.06	0.02											
2.5	−0.05	0.50	0.45	0.22	0.07	0.02	0.01										
3.0	−0.26	0.34	0.49	0.31	0.13	0.04	0.01										
4.0	−0.40	−0.07	0.36	0.43	0.28	0.13	0.05	0.02									
5.0	−0.18	−0.33	0.05	0.36	0.39	0.26	0.13	0.05	0.02								
5.53	0	−0.34	−0.13	0.25	0.40	0.32	0.19	0.09	0.03	0.01							
6.0	0.15	−0.28	−0.24	0.11	0.36	0.36	0.25	0.13	0.06	0.02							
7.0	0.30	0.00	−0.30	−0.17	0.16	0.35	0.34	0.23	0.13	0.06	0.02						
8.0	0.17	0.23	−0.11	−0.29	−0.10	0.19	0.34	0.32	0.22	0.13	0.06	0.03					
8.65	0	0.27	0.06	−0.24	−0.23	0.03	0.26	0.34	0.28	0.18	0.10	0.05	0.02				
9.0	−0.09	0.25	0.14	−0.18	−0.27	−0.06	0.20	0.33	0.31	0.21	0.12	0.06	0.03	0.01			
10.0	−0.25	0.04	0.25	0.06	−0.22	−0.23	−0.01	0.22	0.32	0.29	0.21	0.12	0.06	0.03	0.01		
12.0	0.05	−0.22	−0.08	0.20	0.18	−0.07	−0.24	−0.17	0.05	0.23	0.30	0.27	0.20	0.12	0.07	0.03	0.01

→ No Modulation → No sidebands

when modulation Index ↑ No. of sidebands ↑

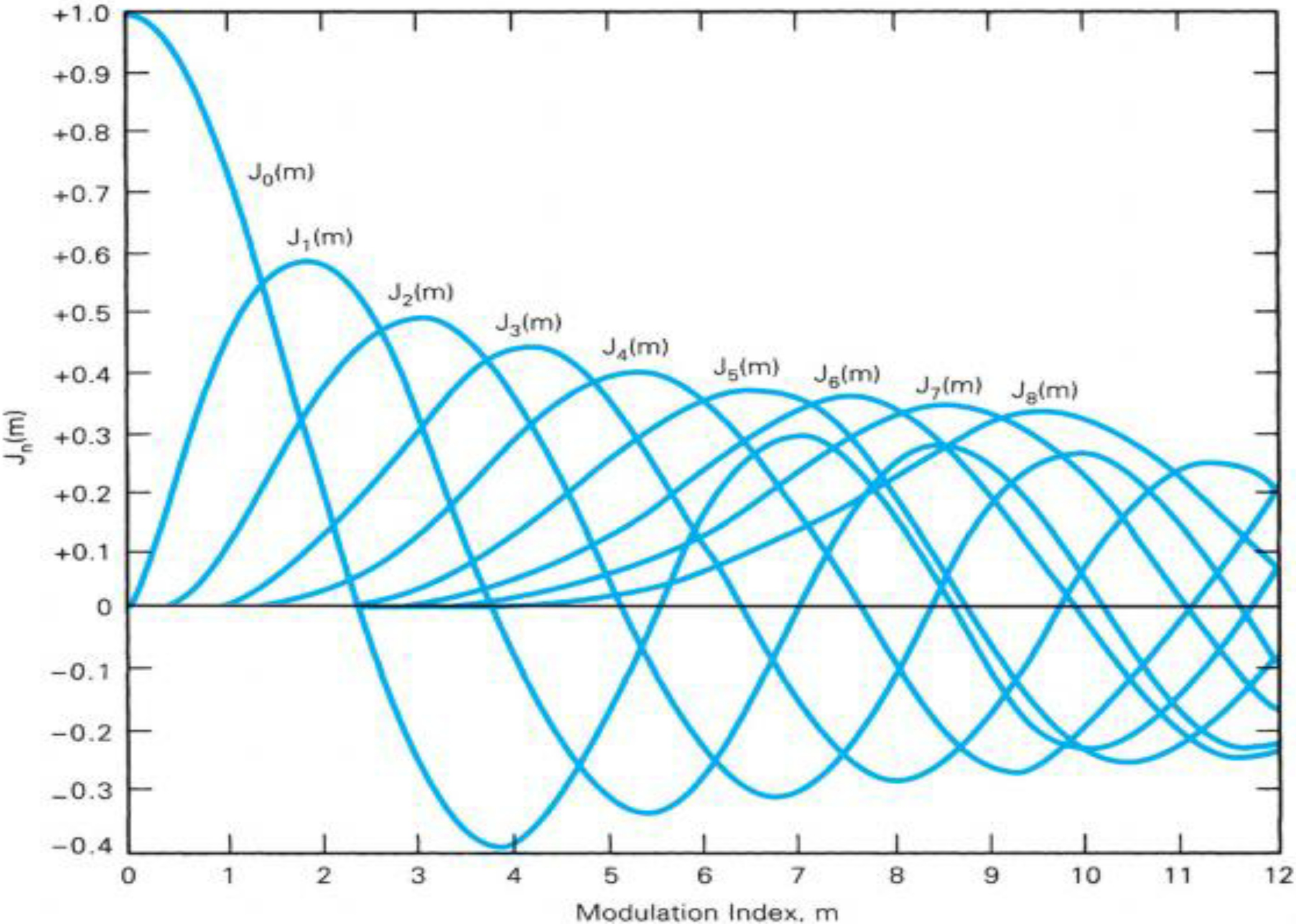
$$\boxed{m_f = \frac{\Delta F}{F_m}}$$

Frequency Spectrum of FM



when modulation increases No. of sideband ↑

Bessel Function



Bandwidth of FM Wave

- Bandwidth of FM wave is a function of modulating signal and modulation index

- **Low index (m<1)** $m_f < 1$ → This signal will act like Am
  - 1 **Bandwidth = 2fm ……Hz** (unit)

- **High index (m>1)** $m_f > 1$
  - 2 **Bandwidth = 2Δf ………Hz**

Bandwidth is dependent on ΔF

Bandwidth of FM Wave

- The actual bandwidth required to pass all significant sideband is two times the product of highest modulating signal frequency

No. of sidebands

3

- **Bandwidth =** $2(n \times f_m)$**.......Hz**

- Carson's rule approximates bandwidth as

4

- **Bandwidth =** $2(\Delta f + f_m)$ **........Hz**

FM is Called Constant Bandwidth System

- Case 1: If $\Delta f = 75$ KHz and fm=500 Hz
  - Bandwidth 1 $= 2(\Delta f + f_m)$ → Carson's Rule
  - **Answer =** 151 KHz ..(1)

- Case 2: If $\Delta f = 75$ KHz and fm=5000 Hz
  - Bandwidth 1 $= 2(\Delta f + f_m)$
  - **Answer =** 160 KHz ..(2)

- Case 3: If $\Delta f = 75$ KHz and fm=10000 Hz
  - Bandwidth 1 $= 2(\Delta f + f_m)$
  - **Answer =** 170 KHz ..(3)

Advantages of FM

- Less noise and interference
- Compared to AM, FM includes low power consumption
- The radiated power is less
- Guard bands separate nearby FM channels
- Lesser geographical interference among adjacent stations

Disadvantages of FM

- High bandwidth
- Receiving area of the FM signal is small
- The antennas for FM systems should be kept close for better communication
- More complicated receiver and transmitter
- Poor spectral efficiency